

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



Le Xuan An

Selective Multi-Teacher Distillation
via Gradient-Alignment Routing
for Multimodal Understanding

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

HÀ NỘI - 2026

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



LE XUAN AN

Selective Multi-Teacher Distillation
via Gradient-Alignment Routing
for Multimodal Understanding

KHOÁ LUẬN TỐT NGHIỆP

Ngành: Khoa học máy tính

Giảng viên hướng dẫn: Tiến sĩ Nguyễn Văn Sơn

Giảng viên đồng hướng dẫn: Giáo sư Võ Đình Hiếu

HÀ NỘI - 2026

VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



Le Xuan An

Selective Multi-Teacher Distillation
via Gradient-Alignment Routing
for Multimodal Understanding

BACHELOR'S THESIS

Major: Computer Science

Advisor: Dr. Nguyen Van Son

Co-Supervisor: Professor Vo Dinh Hieu

HANOI - 2026

Lời cảm ơn

Assurance

I hereby affirm that this thesis is my own original work and has not been submitted previously for any academic or professional award. The thesis was completed through my efforts, with important guidance and support from Dr. Nguyen Van Son and Professor Vo Dinh Hieu, for graduation purposes. I confirm that all information, data and findings presented are original and free from plagiarism. Every source consulted has been properly acknowledged and included in the references. I accept full responsibility for the accuracy and integrity of this thesis in accordance with the regulations of VNU University of Engineering and Technology.

Hanoi, Ngày 31 tháng 3 năm 2026

Student

Le Xuan An

Tóm tắt

Tóm tắt nội dung KL tốt nghiệp: Bài toán, phương pháp giải quyết, kết quả đạt được

Từ khóa: Liệt kê 3-5 từ khoá liên quan

Abstract

Summary of the graduation thesis: The problem statement, the proposed approach and the achieved results.

Keywords: Liệt kê 3-5 từ khoá liên quan

Table of Contents

Lời cảm ơn	
Assurance	i
Tóm tắt	ii
Abstract	iii
Table of Contents	iv
List of Figures	vii
List of Tables	ix
List of Abbreviations	x
Chapter 1 Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement and Research Objectives	3
1.3 Thesis Contributions	4
1.4 Thesis Outline	4
Chapter 2 Preliminaries	5
2.1 Transformer-Based Vision-Language Models	5
2.1.1 Transformer Basics	5
2.1.2 Vision Encoder and Multimodal Fusion	5
2.1.3 Text-Rich Multimodal Understanding	6
2.2 Knowledge Distillation	8
2.2.1 Teacher-Student Learning	8
2.2.2 Temperature-Scaled Output Distributions	8
2.3 Universal Logit Distillation	9

2.3.1	Motivation	9
2.3.2	ULD Formulation	9
2.3.3	Answer-Token Distillation	9
2.4	Multi-Teacher Distillation	10
2.4.1	Per-Teacher KD Loss	10
2.4.2	Need for Selective Routing	10
2.4.3	Learned Routing and Load Balancing	10
2.5	Gradient Alignment	12
2.5.1	Cross-Entropy Gradient Proxy	12
2.5.2	Per-Teacher KD Gradient Proxy	12
2.5.3	Cosine Alignment	12
2.5.4	Alignment Filtering on Top of Gated Routing	13
2.6	Representation Similarity and CKA	14
2.6.1	Why Compare Hidden Layers?	14
2.6.2	Centered Kernel Alignment	14
2.6.3	Connection to Layer-Wise Distillation	14
2.7	Summary	14
Chapter 3 Proposed Method		16
3.1	Problem Formulation	16
3.2	Method Overview	16
3.3	Token-Level Distillation with ULD	17
3.3.1	Supervised Answer Tokens	17
3.3.2	Universal Logit Distillation	18
3.4	Learned Teacher Gate and Load Balancing	19
3.4.1	Student-Conditioned Router	19
3.4.2	Top- k , Capacity, and Fallback	19
3.4.3	Balancing Loss and Expert Bias Update	20
3.5	Gradient Alignment Filtering	21
3.5.1	CE Gradient Proxy	21
3.5.2	KD Gradient Proxy	21
3.5.3	Pooled Cosine Alignment	21
3.5.4	Warmup and Thresholding	21
3.5.5	Alignment Fallback and Effective Weights	22

3.5.6 Sample-Wise KD Aggregation	22
3.6 Final Objective and Training Procedure	23
3.6.1 Final Objective	23
3.6.2 Training Procedure	24
Chapter 4 Evaluation Methodology	25
4.1 Experimental Setup	25
4.2 Datasets	25
4.2.1 Dataset Descriptions	26
4.2.2 Dataset Size and Splits	26
4.3 Evaluation Metrics	26
4.4 Baselines and Compared Methods	28
Chapter 5 Experiment Results	30
5.1 Main Comparison Results	30
5.1.1 Baseline Results	30
5.2 Ablation Study	30
5.3 Qualitative Analysis	30
5.4 Discussion	31
Conclusion and Future Work	32
References	33

List of Figures

1.1	CLIP-style vision-language pretraining aligns image and text representations in a shared embedding space [1, 2].	1
1.2	TinyLLaVA vs. LLaVA-1.5-7B benchmark comparison.	2
1.3	Knowledge distillation transfers information from a teacher model to a smaller student model [3, 4].	2
2.1	Transformer attention block structure [5, 6].	5
2.2	Vision Transformer pipeline with patch embedding and Transformer encoder blocks [7, 8].	6
2.3	Representative architecture families of modern vision-language models.	7
2.4	Representative inputs from the text-rich multimodal benchmarks considered in this thesis [9–11].	7
2.5	Generic teacher-student knowledge distillation setup with hard labels and soft targets [3, 4].	8
2.6	ULD compares ranked probability mass after temperature scaling, sorting, and zero-padding to a common vocabulary length [12]. . . .	9
2.7	Example of sparse routing and expert specialization in a mixture-of-experts architecture [13, 14]. Although our setting is multi-teacher distillation rather than token-level MoE language modeling, the same principles of learned routing, sparse activation, and load balancing are directly relevant.	11
2.8	Illustration of gradient alignment in pooled logit-gradient space. The learned gate first proposes constrained routing weights $\hat{p}_i^{(b)}$; gradient alignment then activates only sufficiently aligned teachers. If none are active, the router falls back to the least-negative teacher.	13
2.9	A high CKA score is not sufficient to conclude that layer-wise transfer will remove the downstream performance gap. Representation similarity and task performance are related but not equivalent.	15

3.1	Overview of the current method. A student-side deep gate proposes sparse teacher weights, top- k and capacity constraints determine routed teachers, a balancing loss regularizes teacher usage, and cosine alignment filters the routed teachers before KD aggregation. Sample superscripts are omitted when clear.	17
3.2	Gate architecture used in the current implementation. The router pools student hidden states over supervised positions, applies Layer-Norm, uses a gated MLP block $\text{SiLU}(\mathbf{g}) \odot \mathbf{v}$, and outputs one routing logit per teacher.	19
4.1	Example image for a TextVQA-style question [15].	26
4.2	Example document image for a DocVQA-style question [16].	27
4.3	Example chart image for a ChartQA-style question [17].	27

List of Tables

4.1	Dataset split sizes (number of QA pairs) for TextVQA, DocVQA and ChartQA [9–11].	28
5.1	Original benchmark results of the two SmolVLM models evaluated with VLM-Eval-Kit.	30

List of Abbreviations

Abbr.	Full Term	Description
ANLS	Average Normalized Levenshtein Similarity	Evaluation metric used for DocVQA.
API	Application Programming Interface	Interface used to query model outputs in black-box settings.
BLIP-2	Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models	A modular vision-language model with a Q-Former bridge.
CE	Cross-Entropy	Supervised loss used for classification or distillation objectives.
ChartQA	Chart Question Answering	Benchmark for question answering over charts.
CLIP	Contrastive Language-Image Pre-training	Dual-encoder vision-language pre-training framework.
CNN	Convolutional Neural Network	Standard visual backbone architecture used as a comparison point to ViT.
DocVQA	Document Visual Question Answering	Benchmark for question answering on document images.
FFN	Feedforward Network	Transformer sublayer applied after attention.
InfoNCE	Information Noise-Contrastive Estimation	Contrastive objective for aligning image and text representations.
JS	Jensen-Shannon	Symmetric divergence measure between probability distributions.
KD	Knowledge Distillation	Transfer of knowledge from a teacher model to a student model.
KL	Kullback-Leibler	Measure used to compare probability distributions.

Abbr.	Full Term	Description
KLD	Kullback-Leibler Divergence	Divergence-based objective widely used in distillation.
LLaVA	Large Language and Vision Assistant	Instruction-tuned projector-based vision-language model.
LLM	Large Language Model	Language model with large parameter scale and strong reasoning ability.
LN	Layer Normalization	Normalization method used to stabilize Transformer training.
MLP	Multi-Layer Perceptron	Feedforward projection block used in Transformers and classifiers.
MMD	Maximum Mean Discrepancy	Similarity metric used for feature-level alignment.
MoE	Mixture of Experts	Sparse architecture that routes tokens to selected experts.
MSA	Multi-Head Self-Attention	Attention block that models token interactions in Transformers.
MTKD	Multi-Teacher Knowledge Distillation	Distillation setting that uses more than one teacher model.
NLP	Natural Language Processing	Field concerned with computational understanding of language.
NLS	Normalized Levenshtein Similarity	String similarity score used in DocVQA evaluation.
OCR	Optical Character Recognition	Extraction of text from images or scanned documents.
PE	Positional Encoding	Representation of token order in Transformer models.
PLN	Parametric Layer Normalization	Layer normalization variant with trainable scale and bias.
Q-Former	Querying Transformer	Bridging module used by BLIP-2 to connect vision and language features.

Abbr.	Full Term	Description
QA	Question Answering	Task of generating answers to natural language questions.
ReLU	Rectified Linear Unit	Common non-linear activation function.
RMSNorm	Root Mean Square Layer Normalization	Lightweight normalization method based on root mean square statistics.
RoPE	Rotary Positional Embedding	Positional encoding method based on rotational transformations.
TextVQA	Text Visual Question Answering	Benchmark for question answering on natural images containing text.
ViT	Vision Transformer	Transformer-based model for image representation learning.
VLM	Vision-Language Model	Model that jointly processes visual and textual inputs.
VLM-Eval-Kit	Vision-Language Model Evaluation Kit	Toolkit used in this thesis to run standardized multimodal evaluation.
VNU	Vietnam National University	Parent university system of VNU University of Engineering and Technology.
VQA	Visual Question Answering	Task of answering questions grounded in visual inputs.

Chapter 1

Introduction

1.1 Background and Motivation

Vision-Language Models (VLMs) jointly process images and text for tasks such as image captioning, visual question answering, document understanding and instruction following [2, 18–21].

In this thesis, *multimodal understanding* refers to practical capabilities such as reading text in images, understanding the input question and performing basic reasoning over visual and textual evidence, rather than broad general intelligence.

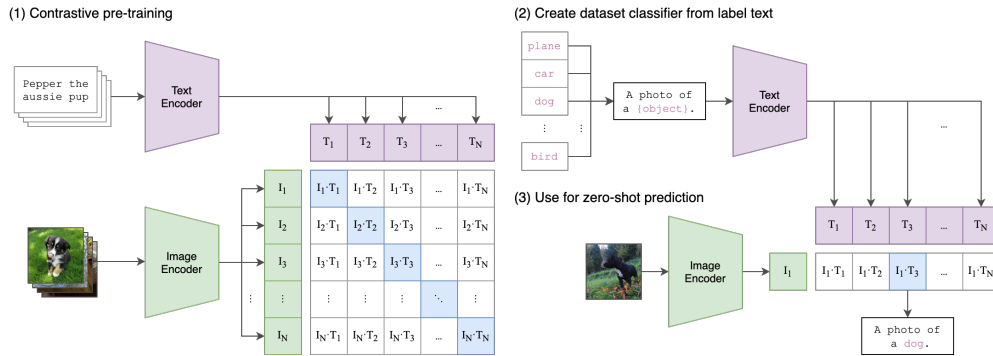


Figure 1.1: CLIP-style vision-language pretraining aligns image and text representations in a shared embedding space [1, 2].

However, high-performing VLMs are often large and expensive in memory, latency and deployment cost, which limits their use in resource-constrained settings [18, 19, 22, 23]. This issue is especially important for text-centric tasks such as scene-text reading, document understanding and chart reasoning [9–11].

Knowledge distillation transfers knowledge from a large teacher to a compact student, achieving competitive performance at lower cost [4, 24]. Figure 1.2 illustrates why this direction matters: compact multimodal models can remain competitive, but closing the gap to stronger models still requires better transfer of knowledge. In the multimodal setting, recent work has also shown that distillation can be used to transfer reasoning or representation knowledge into more efficient VLMs [21, 25].

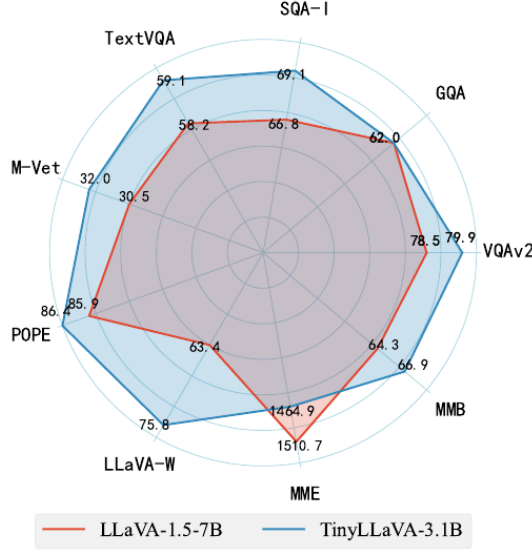


Figure 1.2: Benchmark comparison from TinyLLaVA Figure 1, where a 3.1B small-scale multimodal model remains competitive with the 7B LLaVA-1.5 baseline across several benchmarks [23].

For VLMs, this is especially relevant on text-rich benchmarks such as TextVQA, DocVQA and ChartQA [9–11].

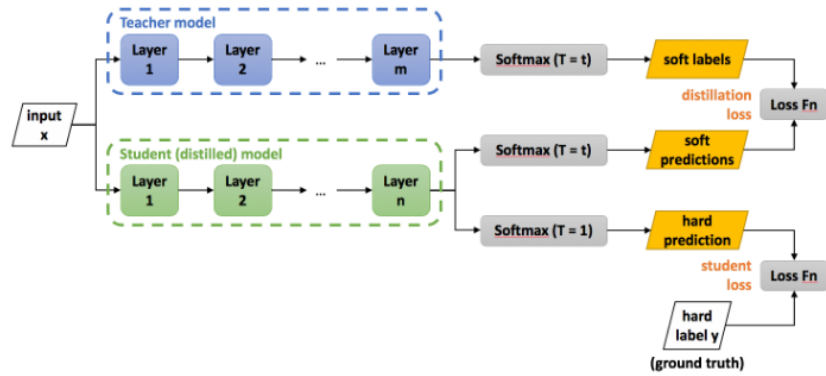


Figure 1.3: Knowledge distillation transfers information from a teacher model to a smaller student model [3, 4].

Since no single teacher excels across all multimodal tasks, *multi-teacher distillation* uses multiple complementary teachers with adaptive selection to provide richer, task-relevant supervision [24, 26, 27].

Beyond improving student performance, this thesis also addresses a meta-level problem: *how to choose the right set of teachers for a specific student*. Prior work

has shown that fixed or uniform teacher assignment is often suboptimal and recent studies have started to explore adaptive teacher weighting or selective teacher transfer [26–28]. **To the best of our knowledge, student-specific teacher-set optimization for multimodal VLM distillation remains a completely unexplored problem in the literature.** In this thesis, teacher selection is treated as part of the learning problem itself rather than as a fixed design choice.

This thesis investigates knowledge distillation for VLMs, focusing on how a compact student can learn from multiple teacher VLMs and how teacher-set optimization improves transfer on text-rich multimodal understanding tasks such as reading image text and solving basic reasoning questions in TextVQA, DocVQA and ChartQA [9–11].

1.2 Problem Statement and Research Objectives

The central problem addressed in this thesis is how to transfer multimodal understanding ability from large VLMs to compact student models without incurring the full cost of large-scale deployment. In particular, we focus on text-rich multimodal tasks where the student must read image text, understand the question and perform basic reasoning. Existing distillation pipelines usually assume either a single teacher or a fixed set of teachers, which leaves open the harder question of how to select the most useful teachers for a given student model.

Given a pool of N teacher VLMs and a fixed compact student, the goal is to select K optimal teachers and train the student to maximize knowledge transfer.

Accordingly, this thesis has three main objectives:

1. formulate multimodal knowledge distillation for compact VLM students in a text-centric benchmark setting,
2. design an adaptive multi-teacher strategy that treats teacher-set selection as part of the learning problem,
3. evaluate whether teacher-set optimization improves transfer for small VLMs on TextVQA, DocVQA and ChartQA.

1.3 Thesis Contributions

The main contributions of this thesis are summarized as follows:

1. We formulate student-specific teacher-set optimization for multimodal VLM distillation, focusing on compact students for text-rich multimodal understanding.
2. We propose an adaptive multi-teacher distillation framework in which teacher selection and knowledge transfer are optimized jointly.
3. We study the method on small VLM students and evaluate it on TextVQA, DocVQA and ChartQA using a unified evaluation pipeline.

1.4 Thesis Outline

The remainder of this thesis is organized as follows:

1. Chapter 2 reviews the background on Transformers, VLMs, knowledge distillation and mixture of experts.
2. Chapter 3 presents the proposed method, including the problem formulation, teacher selection strategy and training procedure.
3. Chapter 4 describes the experimental setup, datasets, evaluation metrics and compared methods.
4. Chapter 5 reports the baseline results, comparison results, ablation studies and discussion.

Chapter 2

Preliminaries

2.1 Transformer-Based Vision-Language Models

2.1.1 Transformer Basics

Modern vision-language models are built on the Transformer architecture [6]. For an input sequence, self-attention maps query, key and value matrices $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ to

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}, \quad (2.1)$$

where d_k is the key dimension. Multi-head attention applies this operation in parallel across several learned subspaces and is followed by a feed-forward network and residual normalization.

For language modeling, the Transformer produces a sequence of hidden states and next-token logits. In this thesis, the student and teacher models are trained or evaluated through their output logits because the distillation method operates at the token-distribution level.

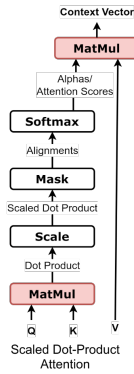


Figure 2.1: Transformer attention block structure [5, 6].

2.1.2 Vision Encoder and Multimodal Fusion

On the visual side, many multimodal systems use a Vision Transformer (ViT) [8], which converts an image into a sequence of patch embeddings and processes

them with Transformer blocks. These visual tokens are then fused with text by a connector module such as cross-attention, a Q-Former, or a learned projector [18–20].

This thesis focuses on compact generative VLMs in the same general family as SmolVLM and related projector-based models. Conceptually, such a model can be written as

$$\mathbf{z} = f_{\theta}(x), \quad (2.2)$$

where x is a multimodal input and $\mathbf{z}$ denotes the output logits used for autoregressive answer generation.

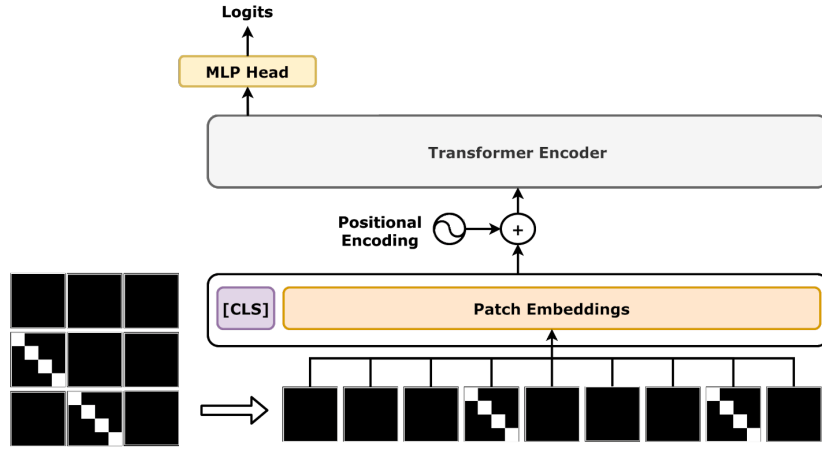
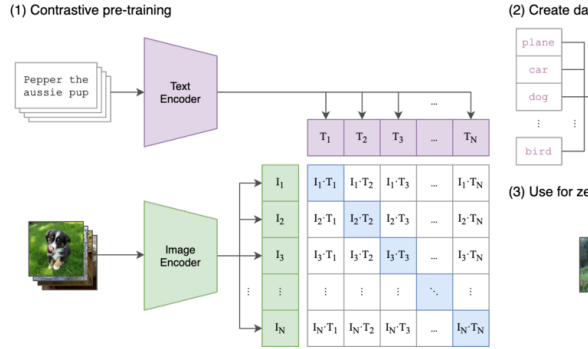


Figure 2.2: Vision Transformer pipeline with patch embedding and Transformer encoder blocks [7, 8].

2.1.3 Text-Rich Multimodal Understanding

The tasks considered in this thesis, namely TextVQA, DocVQA, and ChartQA, require the model to read text in images, understand the question, and perform basic reasoning over visual and textual evidence [9–11]. This setting differs from generic image captioning because success depends strongly on OCR-sensitive recognition and token-level answer generation.



(a) CLIP-style dual encoder [2].

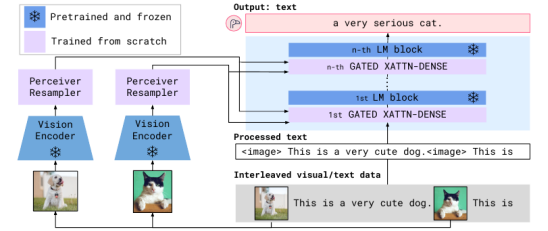
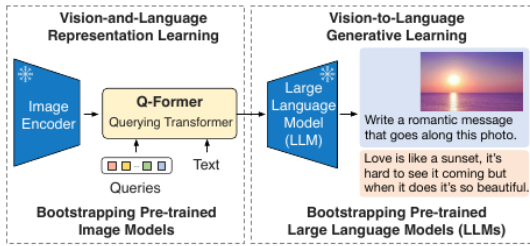


Figure 3: Flamingo architecture overview. Flamingo is a family of visual language models (VLMs) that take as input visual data interleaved with text and produce free-form text as output.

(b) Flamingo cross-attention design [18].



(c) BLIP-2 with Q-Former [19].

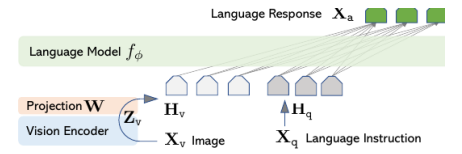


Figure 1: LLaVA network architecture.

(d) LLaVA projector-based design [20].

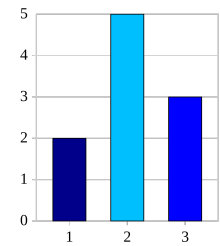
Figure 2.3: Representative architecture families of modern vision-language models.



(a) TextVQA



(b) DocVQA



(c) ChartQA

Figure 2.4: Representative inputs from the text-rich multimodal benchmarks considered in this thesis [9–11].

2.2 Knowledge Distillation

2.2.1 Teacher-Student Learning

Knowledge distillation transfers information from a stronger teacher model to a smaller student model [4, 24]. Let $\mathcal{L}_{CE}$ denote the supervised cross-entropy loss of the student. A generic distillation objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD}, \quad (2.3)$$

where $\mathcal{L}_{KD}$ is a teacher-guided loss and $\alpha \geq 0$ controls its weight.

In token-level distillation, the teacher and student are compared on their output logits rather than only on hard labels. This allows the student to learn richer distributional structure, often called *dark knowledge*.

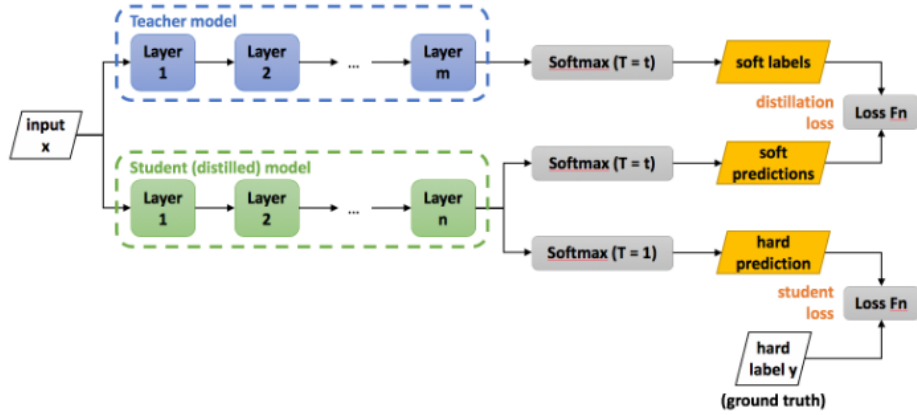


Figure 2.5: Generic teacher-student knowledge distillation setup with hard labels and soft targets [3, 4].

2.2.2 Temperature-Scaled Output Distributions

Let $\mathbf{z}_s$ and $\mathbf{z}_t$ denote student and teacher logits. With temperatures τ_s and τ_t , the corresponding distributions are

$$\mathbf{p}_s = \text{softmax}\left(\frac{\mathbf{z}_s}{\tau_s}\right), \quad \mathbf{p}_t = \text{softmax}\left(\frac{\mathbf{z}_t}{\tau_t}\right). \quad (2.4)$$

Many classical KD methods compare $\mathbf{p}_s$ and $\mathbf{p}_t$ using divergences such as forward KL or Jensen-Shannon divergence. However, such approaches usually assume that teacher and student share the same vocabulary.

2.3 Universal Logit Distillation

2.3.1 Motivation

In this thesis, the teacher pool may contain different VLM families with different tokenizers and vocabulary sizes. Therefore, a KD loss that assumes a shared output space is too restrictive. To address this, we use Universal Logit Distillation (ULD) [12].

2.3.2 ULD Formulation

Given student logits $\mathbf{z}_s \in \mathbb{R}^{V_s}$ and teacher logits $\mathbf{z}_t \in \mathbb{R}^{V_t}$, we first compute their temperature-scaled probability distributions $\mathbf{p}_s$ and $\mathbf{p}_t$. We then sort both vectors in descending order, pad the smaller vocabulary with zeros, and compare the sorted masses with an L_1 distance:

$$\mathcal{L}_{ULD}(\mathbf{z}_s, \mathbf{z}_t) = \|\pi(\mathbf{p}_s) - \pi(\mathbf{p}_t)\|_1, \quad (2.5)$$

where $\pi(\cdot)$ denotes sorting in descending order after zero-padding.

The key advantage of ULD is that it compares distributions by their ranked probability mass rather than by token identity. This makes it directly applicable to cross-tokenizer teacher-student distillation.

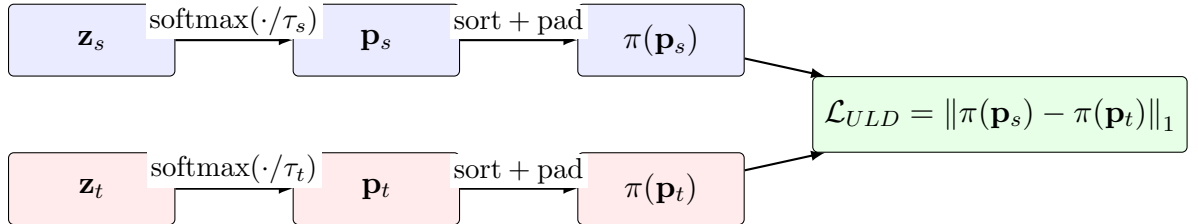


Figure 2.6: ULD compares ranked probability mass after temperature scaling, sorting, and zero-padding to a common vocabulary length [12].

2.3.3 Answer-Token Distillation

For multimodal question answering, distillation should focus on answer tokens rather than prompt tokens. In our implementation, KD is computed only on supervised answer positions after masking ignored tokens, optionally removing EOS,

and truncating student and teacher answer spans to a common length. This gives a token-aligned distillation signal while remaining compatible with different tokenizers.

2.4 Multi-Teacher Distillation

2.4.1 Per-Teacher KD Loss

Let $\mathcal{T} = \{T_1, \dots, T_N\}$ be a pool of frozen teachers. For sample b , teacher T_i produces a per-teacher distillation loss $\mathcal{L}_{KD}^{(b,i)}$. In principle, one could average these losses uniformly,

$$\mathcal{L}_{KD}^{(b)} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{KD}^{(b,i)}, \quad (2.6)$$

but this assumes every teacher is equally useful for every sample.

2.4.2 Need for Selective Routing

In practice, different teachers may provide conflicting guidance. Some teachers may help the supervised objective on a given sample, while others may push the student in a different direction. This motivates a selective routing mechanism that decides, for each sample, which teachers should be active.

2.4.3 Learned Routing and Load Balancing

Recent sparse expert models route each sample or token through only a subset of available experts, which improves efficiency and allows specialized computation [14, 29, 30]. The same high-level idea is useful in multi-teacher distillation: instead of averaging all teachers equally, a learned router can propose teacher weights from student-side features.

Let $\mathbf{h}^{(b)}$ denote a pooled student representation for sample b . A router network $G(\cdot)$ produces gate logits

$$\mathbf{r}^{(b)} = G(\mathbf{h}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\mathbf{r}^{(b)}), \quad (2.7)$$

where $p_i^{(b)}$ is the initial routing weight for teacher T_i .

In sparse routing, these weights are typically refined by additional constraints:

1. **top- k routing**, which keeps only the highest-scoring teachers,
2. **capacity constraints**, which prevent one teacher from receiving too many assignments,
3. **load balancing**, which discourages routing collapse onto a small subset of teachers.

Following the general MoE literature, one can summarize the balancing signal by a soft load $\bar{p}_i$ and a hard assignment load $\bar{q}_i$, and define an auxiliary balancing objective of the form

$$\mathcal{L}_{bal} = N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (2.8)$$

This auxiliary term does not replace the KD objective; instead, it regularizes the router so that different teachers remain usable during training.

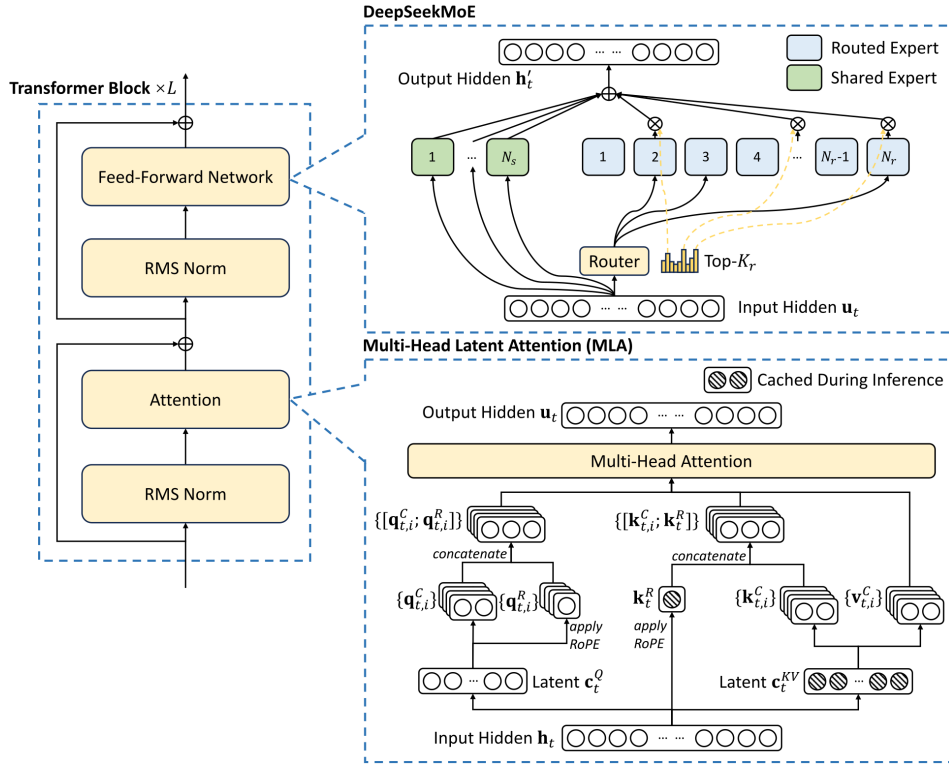


Figure 2.7: Example of sparse routing and expert specialization in a mixture-of-experts architecture [13, 14]. Although our setting is multi-teacher distillation rather than token-level MoE language modeling, the same principles of learned routing, sparse activation, and load balancing are directly relevant.

2.5 Gradient Alignment

2.5.1 Cross-Entropy Gradient Proxy

For sample b , let $\mathcal{L}_{CE}^{(b)}$ be the student cross-entropy loss and let $\mathbf{z}_s^{(b)}$ be the corresponding student logits. We define the cross-entropy logit-gradient proxy as

$$\mathbf{g}_{CE}^{(b)} = \nabla_{\mathbf{z}_s^{(b)}} \mathcal{L}_{CE}^{(b)}. \quad (2.9)$$

This gradient describes the local update direction preferred by supervised learning.

2.5.2 Per-Teacher KD Gradient Proxy

Similarly, for each teacher T_i , we define a KD logit-gradient proxy

$$\mathbf{g}_i^{(b)} = \nabla_{\mathbf{z}_s^{(b)}} \mathcal{L}_{KD}^{(b,i)}. \quad (2.10)$$

This gives a teacher-specific candidate update direction in the same logit space as $\mathbf{g}_{CE}^{(b)}$.

2.5.3 Cosine Alignment

To measure whether teacher T_i is locally consistent with the supervised objective, we compare pooled CE and KD gradient proxies in the same logit space. Let

$$\tilde{\mathbf{g}}_{CE}^{(b)} = \text{Pool}\left(\mathbf{g}_{CE}^{(b)}\right), \quad \tilde{\mathbf{g}}_i^{(b)} = \text{Pool}\left(\mathbf{g}_i^{(b)}\right), \quad (2.11)$$

where $\text{Pool}(\cdot)$ averages over supervised answer positions. The sample-level cosine alignment score is then

$$s_i^{(b)} = \cos\left(\tilde{\mathbf{g}}_{CE}^{(b)}, \tilde{\mathbf{g}}_i^{(b)}\right). \quad (2.12)$$

Thus, $s_i^{(b)}$ is a sample-level agreement score after pooling the supervised-token gradients. A larger value indicates that teacher T_i provides a KD direction more consistent with the supervised objective, while a negative value indicates directional conflict.

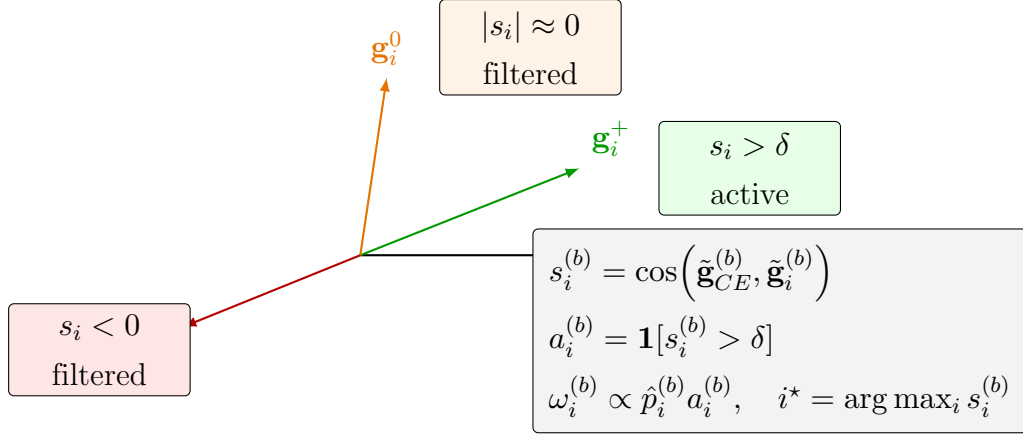


Figure 2.8: Illustration of gradient alignment in pooled logit-gradient space. The learned gate first proposes constrained routing weights $\hat{p}_i^{(b)}$; gradient alignment then activates only sufficiently aligned teachers. If none are active, the router falls back to the least-negative teacher.

2.5.4 Alignment Filtering on Top of Gated Routing

In the current method, alignment does not replace the learned gate. Instead, the gate first proposes constrained routing weights $\hat{p}_i^{(b)}$ after top- k selection and capacity control, and alignment is then used as a binary activation filter:

$$\hat{a}_i^{(b)} = \mathbf{1}[s_i^{(b)} > \delta], \quad (2.13)$$

where δ is an alignment threshold.

If no teacher remains active after thresholding, the method falls back to the least-negative aligned teacher among the currently routed teachers,

$$i^* = \arg \max_i s_i^{(b)}, \quad \hat{a}_i^{(b)} = \mathbf{1}[i = i^*]. \quad (2.14)$$

The final effective teacher weights are then

$$\omega_i^{(b)} = \frac{\hat{p}_i^{(b)} \hat{a}_i^{(b)}}{\sum_{k=1}^N \hat{p}_k^{(b)} \hat{a}_k^{(b)}}. \quad (2.15)$$

This keeps the learned gate responsible for coarse routing and balancing, while alignment acts as a second-stage filter that suppresses routed teachers whose KD gradient conflicts with the supervised objective.

2.6 Representation Similarity and CKA

2.6.1 Why Compare Hidden Layers?

When analyzing whether layer-wise distillation is suitable, it is useful to ask whether a teacher hidden layer and a student hidden layer actually encode similar representations. A common way to study this question is to compare activation matrices collected from the same set of inputs and measure their similarity [31].

2.6.2 Centered Kernel Alignment

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\mathbf{Y} \in \mathbb{R}^{n \times q}$ denote centered activation matrices from two layers over the same n examples. In this thesis, we use linear centered kernel alignment (CKA) as the layer-similarity measure:

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{X}^\top \mathbf{Y}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F}. \quad (2.16)$$

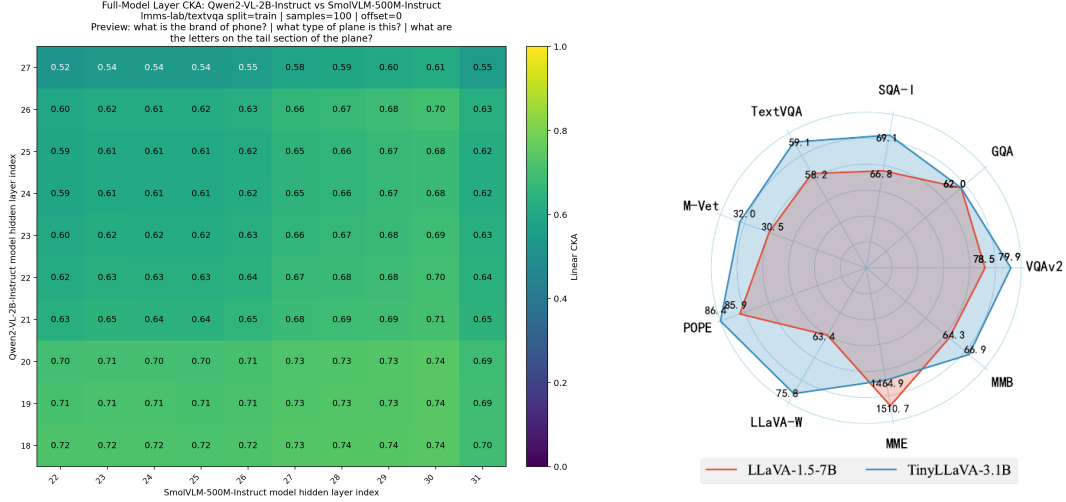
CKA produces a scalar similarity score between two representations. Larger values indicate that the two layers encode more similar geometry across the input set. Compared with earlier similarity measures, CKA is widely used because it is stable in practical settings and is well suited for comparing learned hidden representations across layers and across models [31].

2.6.3 Connection to Layer-Wise Distillation

Later in the thesis, we use CKA as an analysis tool rather than as a training loss. The goal is to examine whether teacher and student layers exhibit clear one-to-one correspondence. If the cross-model layer similarity is weak or diffuse, then directly enforcing layer-wise feature matching is unlikely to be a good inductive bias. This motivates our later conclusion that explicit layer-wise distillation is not effective for the current multimodal distillation setting.

2.7 Summary

The preliminaries in this chapter establish the concepts needed for the proposed method:



(a) High layer similarity in a teacher-student CKA heatmap from our preliminary analysis. (b) Compact VLMs still exhibit a benchmark gap against larger models [23].

Figure 2.9: A high CKA score is not sufficient to conclude that layer-wise transfer will remove the downstream performance gap. Representation similarity and task performance are related but not equivalent.

1. Transformer-based VLMs produce token logits for multimodal answer generation.
2. Knowledge distillation adds a teacher-guided loss to the supervised CE loss.
3. ULD enables output-level distillation even when student and teacher vocabularies differ.
4. Multi-teacher distillation benefits from learned routing, sparse activation, and load balancing when teachers are used unevenly.
5. Gradient alignment provides a principled way to filter routed teachers according to whether their KD update agrees with the supervised objective.

These concepts are combined in Chapter 3 to define the full selective multi-teacher distillation method used in this thesis.

Chapter 3

Proposed Method

3.1 Problem Formulation

We consider text-rich multimodal understanding tasks such as TextVQA, DocVQA, and ChartQA [9–11]. The training set is

$$\mathcal{D} = \{(x_b, y_b)\}_{b=1}^B, \quad (3.1)$$

where x_b is a multimodal input and y_b is the target answer sequence.

Let S_θ denote the student VLM with trainable parameters θ , and let

$$\mathcal{T} = \{T_1, T_2, \dots, T_N\} \quad (3.2)$$

be a pool of frozen teacher VLMs. For each sample b , the student produces logits $\mathbf{z}_s^{(b)}$, and teacher T_i produces teacher logits $\mathbf{z}_t^{(b,i)}$.

The goal of this thesis is not to search for a fixed teacher subset before training. Instead, the student is trained with *alignment-aware gated multi-teacher distillation*: a learned student-side router first proposes sparse teacher weights, balancing terms prevent routing collapse, and gradient alignment then filters the routed teachers according to whether their KD update direction agrees with the supervised cross-entropy objective.

3.2 Method Overview

The proposed method is **alignment-aware gated multi-teacher distillation**. It combines four components:

1. a supervised student objective $\mathcal{L}_{CE}$,
2. a per-teacher logit distillation objective $\mathcal{L}_{KD}^{(b,i)}$,
3. a learned teacher gate with balancing and capacity control,

4. a cosine gradient-alignment filter applied on top of the routed teachers.

At a high level, each teacher first produces a candidate KD signal for the current sample. The student-side gate proposes sparse routing weights from pooled hidden features, balancing terms keep the gate from collapsing, and gradient alignment removes routed teachers whose KD direction is not sufficiently consistent with the supervised objective. The final student update is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD} + \lambda_{bal} \mathcal{L}_{bal}, \quad (3.3)$$

where $\alpha \geq 0$ controls the strength of distillation and λ_{bal} controls the gate-balancing term.

This design is closer in spirit to selective gradient-based distillation [32] and sparse expert routing [29, 30] than to static teacher averaging, but it is adapted here to the multi-teacher multimodal setting with token-level logit distillation.

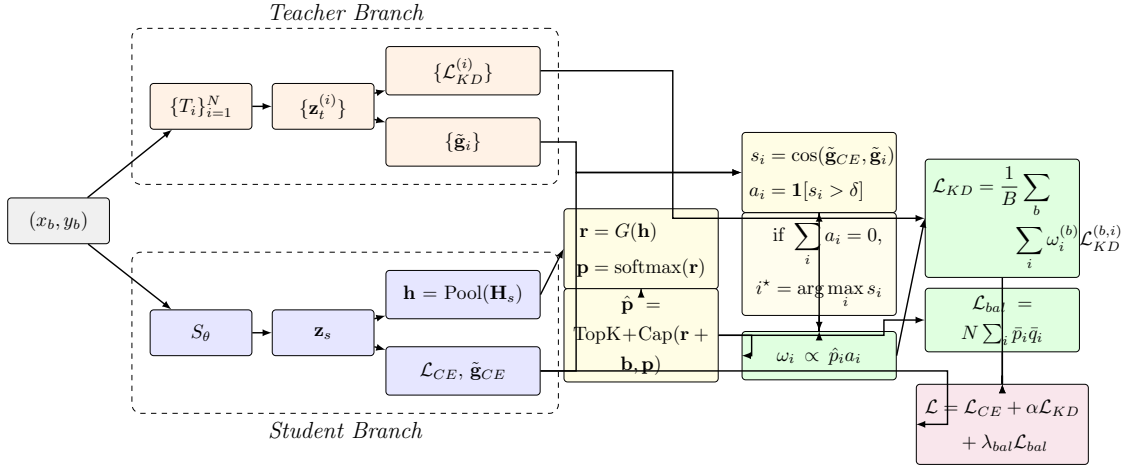


Figure 3.1: Overview of the current method. A student-side deep gate proposes sparse teacher weights, top- k and capacity constraints determine routed teachers, a balancing loss regularizes teacher usage, and cosine alignment filters the routed teachers before KD aggregation. Sample superscripts are omitted when clear.

3.3 Token-Level Distillation with ULD

3.3.1 Supervised Answer Tokens

Distillation is applied only on supervised answer tokens, not on prompt tokens. For sample b , let Ω_b denote the set of supervised token positions after:

1. masking ignored labels,
2. optionally removing the EOS token from both student and teacher answer spans,
3. truncating student and teacher answer spans to a common length.

This produces aligned token sequences for the student and each teacher. The alignment is important because teacher and student models may differ in tokenizer and vocabulary size.

3.3.2 Universal Logit Distillation

For each sample b , token position $j \in \Omega_b$, and teacher T_i , let $\mathbf{z}_{s,j}^{(b)} \in \mathbb{R}^{V_s}$ be the student logits and $\mathbf{z}_{t,j}^{(b,i)} \in \mathbb{R}^{V_i}$ be the teacher logits. The student and teacher probability distributions are

$$\mathbf{p}_{s,j}^{(b)} = \text{softmax}\left(\frac{\mathbf{z}_{s,j}^{(b)}}{\tau_s}\right), \quad \mathbf{p}_{t,j}^{(b,i)} = \text{softmax}\left(\frac{\mathbf{z}_{t,j}^{(b,i)}}{\tau_t}\right), \quad (3.4)$$

where τ_s and τ_t are the student and teacher temperatures.

Following Universal Logit Distillation (ULD) [12], we sort both probability vectors in descending order and pad the smaller vocabulary with zeros. Let $\pi(\cdot)$ denote descending sort after zero-padding to a common size. The token-level ULD loss is

$$\ell_{ULD}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}) = \left\| \pi(\mathbf{p}_{s,j}^{(b)}) - \pi(\mathbf{p}_{t,j}^{(b,i)}) \right\|_1. \quad (3.5)$$

The per-sample, per-teacher KD loss is the mean over supervised answer tokens:

$$\mathcal{L}_{KD}^{(b,i)} = \frac{1}{|\Omega_b|} \sum_{j \in \Omega_b} \ell_{ULD}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}). \quad (3.6)$$

ULD is particularly suitable here because it does not require the student and teacher to share the same vocabulary. This is important in our setting since the teacher pool may contain VLMs from different model families.

3.4 Learned Teacher Gate and Load Balancing

3.4.1 Student-Conditioned Router

When the teacher pool contains multiple models, the method first builds a student-conditioned router instead of averaging teachers uniformly. Let $\mathbf{H}_s^{(b)}$ denote the student hidden states captured at the routing source layer for sample b . We pool these hidden states over supervised answer positions to obtain a routing feature

$$\mathbf{h}^{(b)} = \text{Pool}\left(\mathbf{H}_s^{(b)}\right). \quad (3.7)$$

A deep router $G(\cdot)$ then produces gate logits

$$\mathbf{r}^{(b)} = G\left(\mathbf{h}^{(b)}\right), \quad \mathbf{p}^{(b)} = \text{softmax}\left(\mathbf{r}^{(b)}\right), \quad (3.8)$$

where $p_i^{(b)}$ is the initial gate weight for teacher T_i .

The implemented gate is a compact gated MLP applied to pooled student features. As illustrated in Figure 3.2, the router first normalizes $\mathbf{h}^{(b)}$, projects it to a hidden state twice the router width, splits the result into gate and value branches, applies a SiLU gate, and finally projects to N teacher logits.

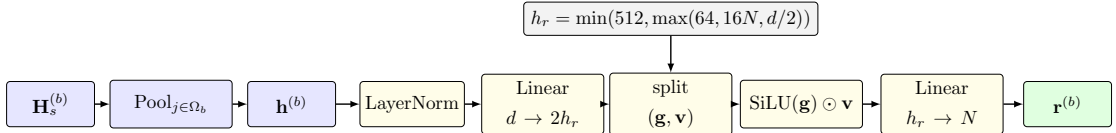


Figure 3.2: Gate architecture used in the current implementation. The router pools student hidden states over supervised positions, applies LayerNorm, uses a gated MLP block $\text{SiLU}(\mathbf{g}) \odot \mathbf{v}$, and outputs one routing logit per teacher.

3.4.2 Top- k , Capacity, and Fallback

The router is made sparse by applying top- k routing to bias-adjusted scores

$$\tilde{r}_i^{(b)} = r_i^{(b)} + b_i, \quad (3.9)$$

where b_i is a teacher-specific expert bias. Let $\mathcal{K}_b = \text{TopK}(\tilde{\mathbf{r}}^{(b)}, k)$ be the candidate teacher set for sample b . To prevent one teacher from receiving too many assignments, each teacher is additionally limited by a capacity

$$C = \left\lceil \phi \frac{Bk}{N} \right\rceil, \quad (3.10)$$

where B is the mini-batch size, N is the number of teachers, and $\phi > 0$ is the capacity factor.

Let $m_i^{(b)} \in \{0, 1\}$ denote the resulting assignment mask after top- k selection and capacity pruning. The constrained gate weights are

$$\hat{p}_i^{(b)} = \frac{m_i^{(b)} p_i^{(b)}}{\sum_{j=1}^N m_j^{(b)} p_j^{(b)}}. \quad (3.11)$$

If capacity pruning leaves a sample without any assigned teacher, the algorithm falls back to the teacher with the largest bias-adjusted routing score $\tilde{r}_i^{(b)}$, ensuring that at least one teacher remains active before alignment filtering.

3.4.3 Balancing Loss and Expert Bias Update

The gate is regularized by a balancing loss that uses both soft routing weights and hard assignments. Define the soft load

$$\bar{p}_i = \frac{1}{B} \sum_{b=1}^B p_i^{(b)} \quad (3.12)$$

and the hard assignment load

$$\bar{q}_i = \frac{1}{Bk} \sum_{b=1}^B \sum_{j=1}^k \mathbf{1}[\text{TopK}_j(\tilde{\mathbf{r}}^{(b)}) = i]. \quad (3.13)$$

The balancing objective is

$$\mathcal{L}_{bal} = N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (3.14)$$

In addition, the expert bias is updated from the per-batch assignment load ℓ_i to discourage teacher overuse:

$$b_i \leftarrow b_i - \eta_b \left(\ell_i - \frac{1}{N} \sum_{j=1}^N \ell_j \right), \quad (3.15)$$

where η_b is a small bias-update rate.

3.5 Gradient Alignment Filtering

3.5.1 CE Gradient Proxy

For each sample b , let $\mathcal{L}_{CE}^{(b)}$ be the student cross-entropy loss on supervised answer tokens. We define the CE logit-gradient proxy as

$$\mathbf{g}_{CE}^{(b)} = \nabla_{\mathbf{z}_s^{(b)}} \mathcal{L}_{CE}^{(b)}. \quad (3.16)$$

In implementation, this gradient is obtained in closed form from the student probability distribution and target labels, which avoids an additional backward pass through the full model.

3.5.2 KD Gradient Proxy

For each teacher T_i , the KD logit-gradient proxy is

$$\mathbf{g}_i^{(b)} = \nabla_{\mathbf{z}_s^{(b)}} \mathcal{L}_{KD}^{(b,i)}. \quad (3.17)$$

This proxy is computed directly from the selected distillation loss. In the default setting of this thesis, $\mathcal{L}_{KD}^{(b,i)}$ is the ULD loss in Equation 3.6.

3.5.3 Pooled Cosine Alignment

For each sample b and teacher T_i , we measure directional agreement between supervised learning and KD by cosine similarity:

$$\tilde{\mathbf{g}}_{CE}^{(b)} = \text{Pool}\left(\mathbf{g}_{CE}^{(b)}\right), \quad \tilde{\mathbf{g}}_i^{(b)} = \text{Pool}\left(\mathbf{g}_i^{(b)}\right), \quad (3.18)$$

$$s_i^{(b)} = \cos\left(\tilde{\mathbf{g}}_{CE}^{(b)}, \tilde{\mathbf{g}}_i^{(b)}\right). \quad (3.19)$$

Here, $\text{Pool}(\cdot)$ averages the logit-gradient tensor over supervised answer positions, producing one pooled CE gradient and one pooled KD gradient per teacher. This is the quantity used by the current implementation.

3.5.4 Warmup and Thresholding

The alignment filter is enabled only after an optional warmup phase. Let $\rho \in [0, 1]$ denote the warmup ratio with respect to the total number of training steps. During

the first ρ fraction of training, the method uses only the constrained gate weights $\hat{p}_i^{(b)}$. After warmup, a teacher is considered alignment-active when

$$a_i^{(b)} = \mathbf{1}[s_i^{(b)} > \delta], \quad (3.20)$$

where δ is the gradient-alignment threshold.

3.5.5 Alignment Fallback and Effective Weights

Alignment is applied on top of the already routed teachers. If at least one routed teacher satisfies Equation 3.20, the final effective teacher weights are

$$w_i^{(b)} = \frac{\hat{p}_i^{(b)} a_i^{(b)}}{\sum_{j=1}^N \hat{p}_j^{(b)} a_j^{(b)}}. \quad (3.21)$$

If none of the routed teachers exceeds the alignment threshold, the method falls back to the routed teacher with the largest alignment score:

$$i^* = \arg \max_{i: \hat{p}_i^{(b)} > 0} s_i^{(b)}, \quad a_i^{(b)} = \mathbf{1}[i = i^*]. \quad (3.22)$$

Equation 3.21 is then recomputed with this fallback mask. As a result, the current implementation never drops KD entirely for a sample after routing; instead, it keeps the least-negative routed teacher when necessary.

3.5.6 Sample-Wise KD Aggregation

The routed KD loss for sample b is the weighted combination

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}. \quad (3.23)$$

The batch-level KD loss is then the mean over samples:

$$\mathcal{L}_{KD} = \frac{1}{B} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)}. \quad (3.24)$$

This rule combines two stages: the gate proposes sparse teacher weights, and gradient alignment filters the routed teachers before the final weighted KD average is formed.

3.6 Final Objective and Training Procedure

3.6.1 Final Objective

The final optimization objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD} + \lambda_{bal} \mathcal{L}_{bal}, \quad (3.25)$$

where $\mathcal{L}_{CE}$ is the standard supervised next-token cross-entropy loss and $\mathcal{L}_{KD}$ is defined in Equation 3.24. The balance term $\mathcal{L}_{bal}$ is defined in Equation 3.14.

Equation 3.25 has an intuitive interpretation:

- $\mathcal{L}_{CE}$ ensures the student always follows the ground-truth supervision,
- $\alpha \mathcal{L}_{KD}$ adds teacher guidance through gated and alignment-filtered multi-teacher routing,
- $\lambda_{bal} \mathcal{L}_{bal}$ regularizes the teacher gate against routing collapse.

3.6.2 Training Procedure

Algorithm 1 Alignment-aware gated multi-teacher distillation

Require: mini-batch $\{(x_b, y_b)\}_{b=1}^B$, frozen teachers $\{T_i\}_{i=1}^N$

- 1: Run S_θ on the batch to obtain $\mathbf{z}_s$, $\mathbf{H}_s$, and $\mathcal{L}_{CE}$
 - 2: Compute $\mathbf{r} = G(\text{Pool}(\mathbf{H}_s))$ and $\mathbf{p} = \text{softmax}(\mathbf{r})$
 - 3: Apply top- k , capacity, and routing fallback to obtain $\hat{\mathbf{p}}$
 - 4: **for** each teacher T_i **do**
 - 5: Compute teacher logits $\mathbf{z}_t^{(i)}$
 - 6: Compute $\mathcal{L}_{KD}^{(b,i)}$ and pooled KD gradient $\tilde{\mathbf{g}}_i^{(b)}$
 - 7: **end for**
 - 8: Compute pooled CE gradient $\tilde{\mathbf{g}}_{CE}^{(b)}$ and alignment scores $s_i^{(b)}$
 - 9: **if** warmup is finished **then**
 - 10: Set $a_i^{(b)} = \mathbf{1}[s_i^{(b)} > \delta]$
 - 11: **if** $\sum_i a_i^{(b)} = 0$ **then**
 - 12: Activate $i^* = \arg \max_{i: \hat{p}_i^{(b)} > 0} s_i^{(b)}$
 - 13: **end if**
 - 14: Compute effective weights $w_i^{(b)} \propto \hat{p}_i^{(b)} a_i^{(b)}$
 - 15: **else**
 - 16: Use routed weights $w_i^{(b)} = \hat{p}_i^{(b)}$
 - 17: **end if**
 - 18: Compute $\mathcal{L}_{KD}$, $\mathcal{L}_{bal}$, and $\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD} + \lambda_{bal} \mathcal{L}_{bal}$
 - 19: Update student parameters θ ; keep all teachers frozen; update gate bias from assignment load
-

Compared with static multi-teacher averaging, this procedure makes the teacher contribution adaptive at the sample level. Compared with fixed teacher-set optimization, it avoids committing to one teacher subset for the entire training run. As a result, the method can exploit complementary teachers while still biasing the update toward teachers whose current KD signal is closer to the supervised objective and toward gates that remain balanced across teachers.

Chapter 4

Evaluation Methodology

4.1 Experimental Setup

All experiments follow a shared evaluation protocol for fair comparison across methods. The study focuses on the two compact student models, **SmolVLM-500M** and **SmolVLM-256M**, evaluated on text-rich multimodal benchmarks. All model evaluations are conducted using **VLM-Eval-Kit** [33]. In brief, the toolkit works as follows:

1. Load a benchmark split and apply its prompt template or input format.
2. Run model inference through a unified model wrapper interface.
3. Post-process raw generations into normalized answers.
4. Compute dataset-specific scores (e.g., VQA Accuracy, ANLS, Relaxed Accuracy) and aggregate final results.

This unified pipeline ensures a consistent evaluation procedure across all compared models.

4.2 Datasets

We evaluate on three text-centric VQA benchmarks: **TextVQA**, **DocVQA** and **ChartQA**. The main purpose is to measure whether the model can **read textual content in images, understand questions and perform basic reasoning** across different visual domains (natural scenes, scanned documents and charts). These datasets are selected because they require OCR-heavy understanding, cover complementary visual domains and include retrieval, layout understanding and numerical reasoning.

4.2.1 Dataset Descriptions

TextVQA [9]: evaluates question answering on natural images containing scene text. It focuses on reading text in-the-wild and combining it with question understanding. *Sample question*: “What word is written on the stop sign?” *Sample answer*: “No word is visible on this sign.”



Figure 4.1: Example image for a TextVQA-style question [15].

DocVQA [10]: evaluates VQA on real document images (scans/photos), where layout, paragraphs, tables and forms are important. It emphasizes document structure understanding in addition to OCR. *Sample question*: “What is the invoice number shown in this document?” *Sample answer*: “4572.”

ChartQA [11]: evaluates question answering over charts. It tests chart reading plus numerical and logical reasoning to derive the final answer. *Sample question*: “Which year has the highest sales value in the chart?” *Sample answer*: “The second (middle) bar has the highest value.”

4.2.2 Dataset Size and Splits

4.3 Evaluation Metrics

We report benchmark-specific official metrics for each dataset:

- **TextVQA: VQA Accuracy**. For a predicted answer $\hat{a}$, the per-question score



Figure 4.2: Example document image for a DocVQA-style question [16].

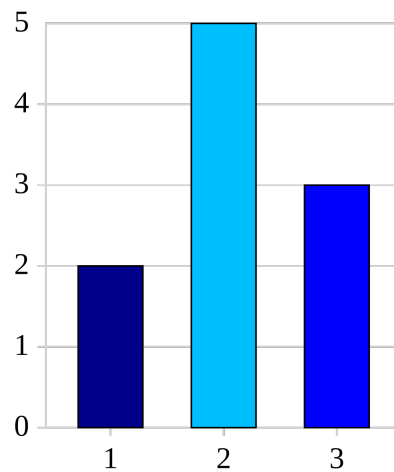


Figure 4.3: Example chart image for a ChartQA-style question [17].

Dataset	Train	Val	Test	Total
TextVQA	34,602	5,000	5,734	45,336
DocVQA	39,463	5,349	5,188	50,000
ChartQA-H (human)	7,398	960	1,250	9,608
ChartQA-M (machine)	20,901	960	1,250	23,111
ChartQA (H+M, combined)	28,299	1,920	2,500	32,719

Table 4.1: Dataset split sizes (number of QA pairs) for TextVQA, DocVQA and ChartQA [9–11].

is

$$s_{\text{VQA}} = \min \left(\frac{\# \text{annotators that provided } \hat{a}}{3}, 1 \right),$$

and the final score is the mean over all questions [9].

- **DocVQA: ANLS** (Average Normalized Levenshtein Similarity):

$$\text{NLS}(a, \hat{a}) = 1 - \frac{d_{\text{Lev}}(a, \hat{a})}{\max(|a|, |\hat{a}|)}, \quad s_{\text{ANLS}} = \begin{cases} \text{NLS}(a, \hat{a}), & \text{NLS} \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

The final ANLS is the average score over all samples [10].

- **ChartQA: Relaxed Accuracy.** A prediction is correct if it matches exactly for text answers, or if the relative error is within 5% for numeric answers:

$$\frac{|\hat{y} - y|}{|y|} \leq 0.05.$$

The final score is the proportion of correct predictions [11].

4.4 Baselines and Compared Methods

Full finetuning. This baseline updates all trainable parameters of the student VLM on the target training data. It provides a strong upper-baseline adaptation setting because the model is optimized directly for the downstream task without distillation constraints. The training objective is the standard supervised next-token prediction loss:

$$\mathcal{L}_{\text{FT}} = - \sum_{t=1}^T \log p_{\theta}(y_t \mid x, y_{<t}), \quad (4.1)$$

where x denotes the multimodal input, y_t is the target token at step t and θ denotes all trainable parameters of the model.

Chapter 5

Experiment Results

5.1 Main Comparison Results

This section compares the proposed method with the baselines on TextVQA, DocVQA and ChartQA.

5.1.1 Baseline Results

Before applying any adaptation or distillation, we report the original performance of the two base student models, **SmolVLM-500M** and **SmolVLM-256M**. These baseline results establish the starting point for later comparison. All results are obtained using **VLM-Eval-Kit** under the same evaluation pipeline described in Chapter 4.

Model	TextVQA	DocVQA	ChartQA
SmolVLM-500M	56.75	59.5247	62.8
SmolVLM-256M	47.17	48.6411	45.04

Table 5.1: Original benchmark results of the two SmolVLM models evaluated with VLM-Eval-Kit.

5.2 Ablation Study

This section analyzes the effect of teacher-set selection, weighting strategy and student model size on final performance.

5.3 Qualitative Analysis

This section presents representative examples to inspect how distillation affects text reading and basic reasoning behavior.

5.4 Discussion

This section discusses the trade-off between model size, teacher supervision and transfer performance across the evaluated benchmarks.

Conclusion and Future Work

References

- [1] Wikimedia Commons contributors. File: Contrastive language-image pretraining.png. Wikimedia Commons, 2024. MIT License. Accessed: 2026-03-06.
- [2] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021.
- [3] Wikimedia Commons contributors. File: Knowledge-distillation-example.png. Wikimedia Commons, 2023. CC0 1.0. Source: Intel Distiller knowledge distillation page. Accessed: 2026-03-07.
- [4] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network, 2015.
- [5] Wikimedia Commons contributors. File: Transformer, attention block diagram.png. Wikimedia Commons, 2024. CC BY 4.0. Accessed: 2026-03-06.
- [6] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, pages 5998–6008, 2017.
- [7] Wikimedia Commons contributors. File: Vision transformer.png. Wikimedia Commons, 2024. CC BY 4.0. Accessed: 2026-03-06.
- [8] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021.
- [9] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. Towards vqa models that can read, 2019.
- [10] Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. Docvqa: A dataset for vqa on document images, 2021.

- [11] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning, 2022.
- [12] Nicolas Boizard, Kevin El Haddad, Pierre Colombo, and Chloé Clavel. Towards cross-tokenizer distillation: The universal logit distillation loss for llms, 2024.
- [13] Wikimedia Commons contributors. File: Deepseek moe and mla (deepseek-v2).svg. Wikimedia Commons, 2025. MIT License. Accessed: 2026-03-06.
- [14] DeepSeek-AI. Deepseek-v2: A strong, economical, and efficient mixture-of-experts language model, 2024.
- [15] Wikimedia Commons contributors. File: Stop sign.jpg. Wikimedia Commons, 2006. Public domain. Accessed: 2026-03-06.
- [16] Wikimedia Commons contributors. File: Receiptswiss.jpg. Wikimedia Commons, 2015. CC BY-SA 3.0. Accessed: 2026-03-06.
- [17] Wikimedia Commons contributors. File: Bar chart.svg. Wikimedia Commons, 2016. CC BY-SA 4.0. Accessed: 2026-03-06.
- [18] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning, 2022.
- [19] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models, 2023.
- [20] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning, 2023.
- [21] Yushi Hu, Otilia Stretcu, Chun-Ta Lu, Krishnamurthy Viswanathan, Kenji Hata, Enming Luo, Ranjay Krishna, and Ariel Fuxman. Visual program distillation: Distilling tools and programmatic reasoning into vision-language models, 2024.

- [22] Xiangxiang Chu, Limeng Qiao, Xinyang Lin, Shuang Xu, Yang Yang, Yiming Hu, Fei Wei, Xinyu Zhang, Bo Zhang, Xiaolin Wei, and Chunhua Shen. Mobilevlm: A fast, strong and open vision language assistant for mobile devices, 2023.
- [23] Baichuan Zhou, Ying Hu, Xi Weng, Junlong Jia, Jie Luo, Xien Liu, Ji Wu, and Lei Huang. Tinyllava: A framework of small-scale large multimodal models, 2024.
- [24] Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao, and Tianyi Zhou. A survey on knowledge distillation of large language models, 2024.
- [25] Kaicheng Yang, Tiancheng Gu, Xiang An, Haiqiang Jiang, Xiangzi Dai, Ziyong Feng, Weidong Cai, and Jiankang Deng. Clip-cid: Efficient clip distillation via cluster-instance discrimination, 2024.
- [26] Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, Ming Gong, Yan Fu, and Daxin Jiang. Reinforced multi-teacher selection for knowledge distillation, 2020.
- [27] Hailin Zhang, Defang Chen, and Can Wang. Adaptive multi-teacher knowledge distillation with meta-learning, 2023.
- [28] Yu-Chu Yu, Chi-Pin Huang, Jr-Jen Chen, Kai-Po Chang, Yung-Hsuan Lai, Fu-En Yang, and Yu-Chiang Frank Wang. Select and distill: Selective dual-teacher knowledge transfer for continual learning on vision-language models, 2024.
- [29] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, 2017.
- [30] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022.
- [31] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3519–3529. PMLR, 2019.

- [32] Yichen Zhu and Yi Wang. Student customized knowledge distillation: Bridging the gap between student and teacher. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5057–5066, 2021.
- [33] Haodong Duan, Xinyu Fang, Junming Yang, et al. Vlmevalkit: An open-source toolkit for evaluating large multi-modality models, 2024.